

Third Generation Sequencing

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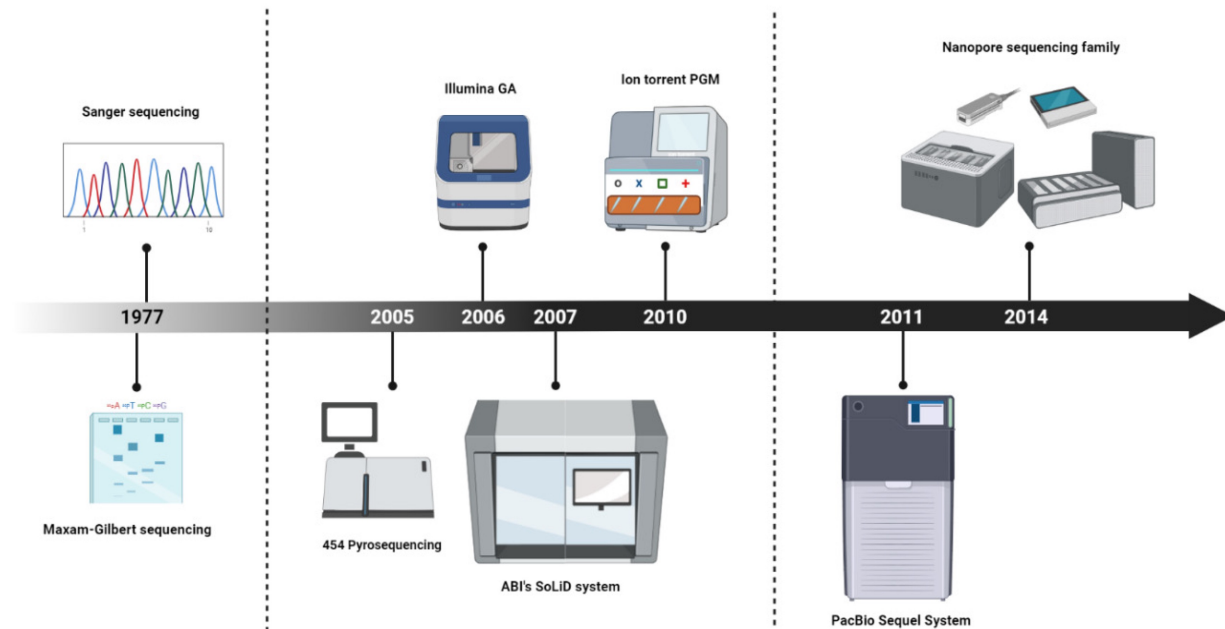
What is Third Generation Sequencing?

Whilst Illumina sequencing is highly accurate and inexpensive, several challenges emerged using short-read data:

- Difficulty assembling genomes
- Unable to resolve genome complexities
- Resource requirements

Third Generation Sequencing includes **Oxford Nanopore Technology (ONT)** and **Pacific Biosciences sequencing** that can sequence long DNA fragments in a single read strived to overcome such limitations. Generally, they can:

- Sequence longer reads
- Perform real-time sequencing
- Single-molecule sequencing



Third Generation Sequencing Platforms

	Second Generation sequencing Illumina NovaSeq X	PacBio: Single Molecule Real-Time Sequencing SMRT: Vega	Oxford Nanopore Technology: MinION	Oxford Nanopore Technology: PromethION
Read length (bp)	300 max (600 paired)	15,000-25000 (up to 100,000 for HiFi)	90,000	33,000-270,000
Ave yield (Gb)	1600-2000	60	25-50	20-130
Cost Per Run (\$)	2500	850	~800	~850
Hardware Cost (\$)	1,250,000	169,000	1000	25,000
Observed Error Rate	<1%	<1%-5%	<1%-5%	<1%-5%



Nanopore Technology

PromethION



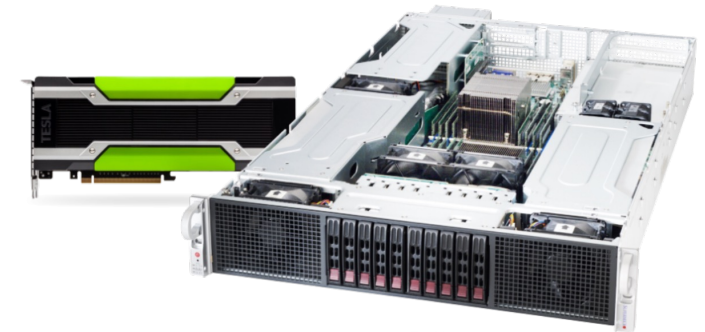
GridION



MinION

Flongle

**HPC + GPU
(Nvidia > 1080)**



Nanopore Sequencing Technology

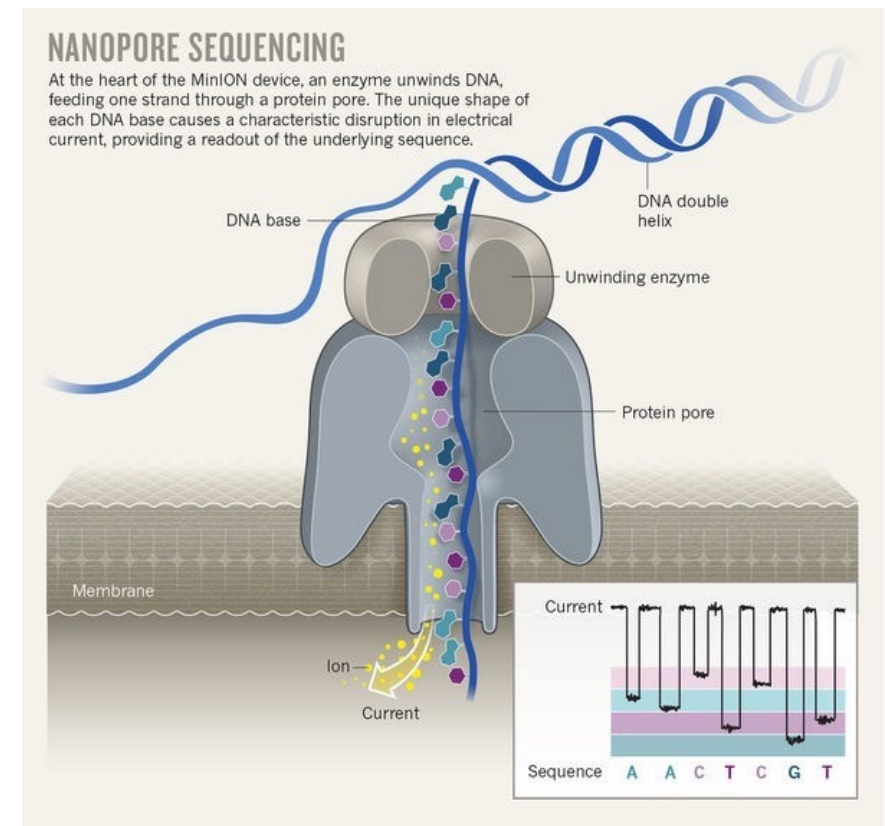
“Portable, real-time sequencing analyses”

ONT is a long-read sequencing platform, which offers several **advantages** over short-read sequencers such as Illumina:

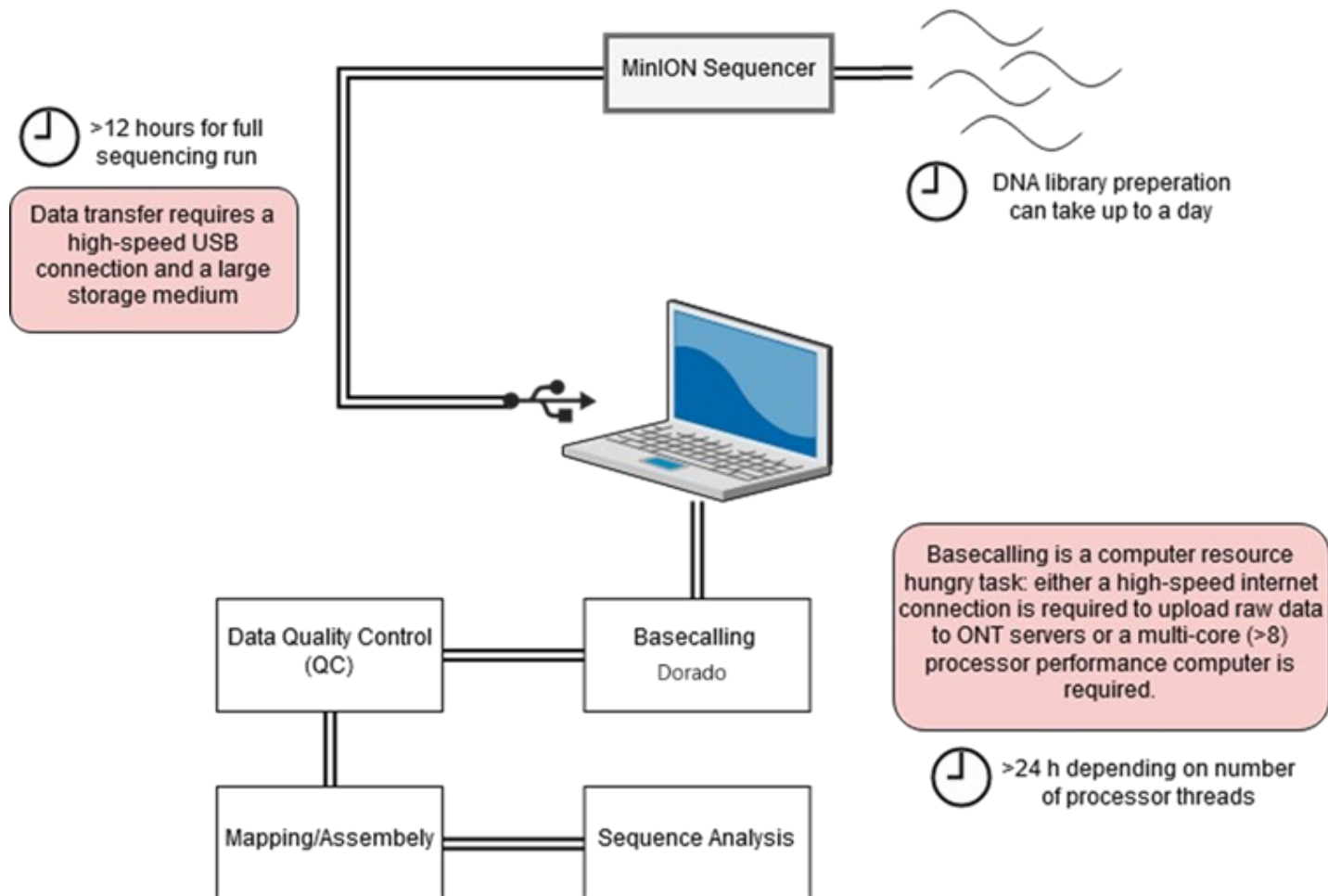
- Can span **repetitive regions**
- Minimal **sample preparation**
- Can be useful for **real-time sequencing** (in the field)
- Resolve **complex genomic structures**
- Detection of **epigenetic modifications** (Methylation)

However, there are still some **limitations**:

- **Lower accuracy/ higher error rate**
- Basecalling large amounts of data requires a **GPU resource**
- High dependency on **community testing** to address issues with the platform
- **Adapting existing pipelines** and tools for ONT data

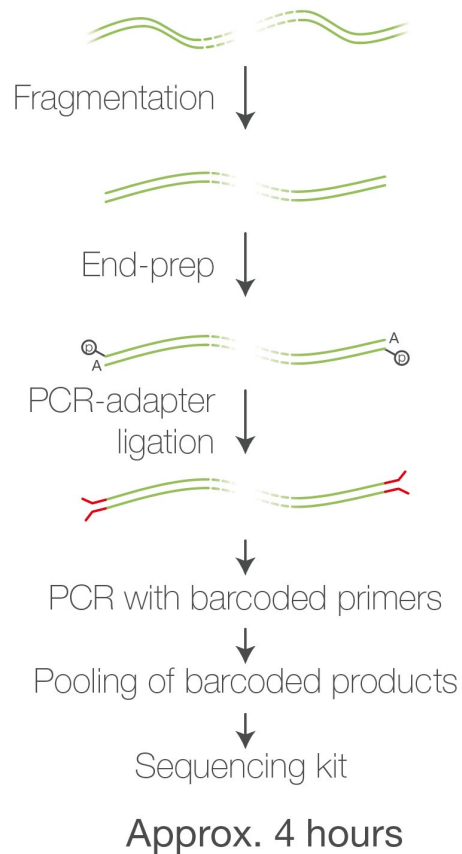


Sequencing Pipeline

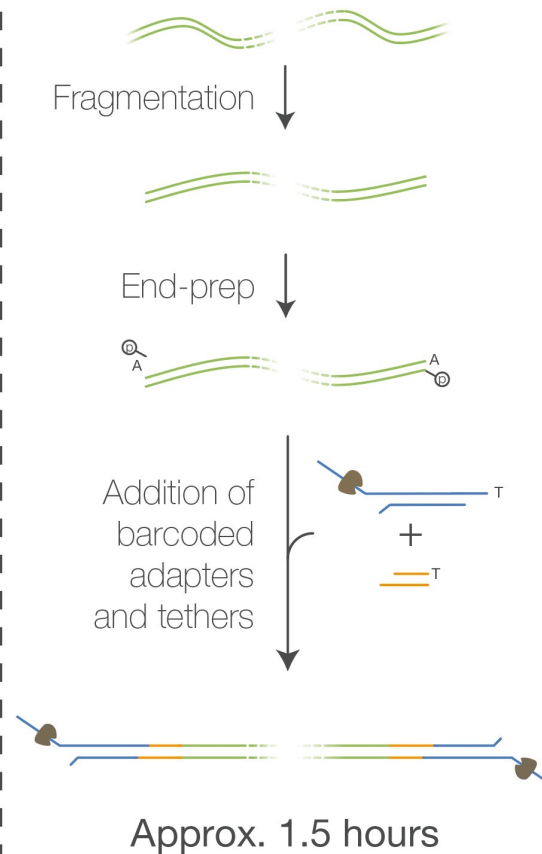


Library preparation and loading

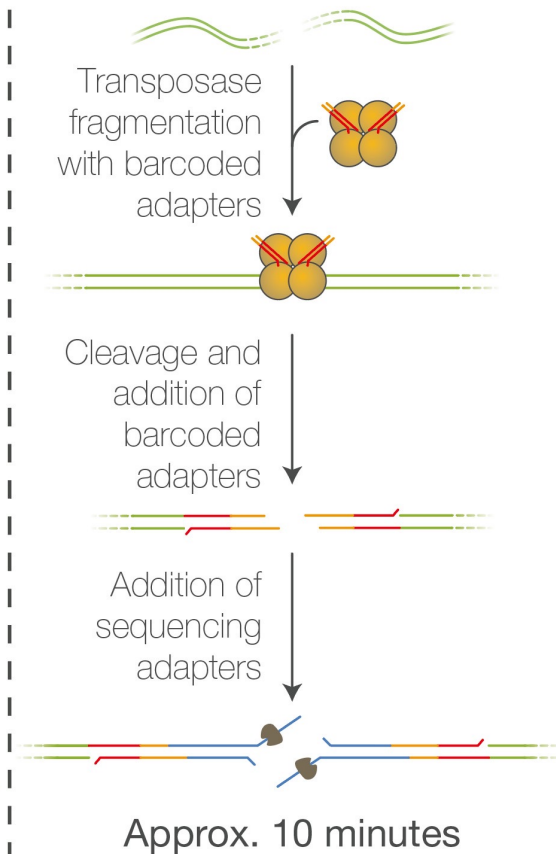
a) PCR barcoding



b) PCR-free barcoding



c) Rapid barcoding



Flow cells and chemistries

Flow Cells

- A flow cell is a key component of the ONT platform where the DNA or RNA samples are loaded for sequencing.
- The primary flow cells available are R9.4.1 and R10.4.1.

Chemistries

- Chemistries refer to the set of reagents and protocols used to prepare the sample for sequencing, influencing the quality and accuracy of the sequencing reads. There are different types.

Kits

- Kits are the complete sets of materials provided for sequencing specific types of samples (DNA, RNA) using particular chemistries.
- Each kit includes the required reagents, primers, and protocols to ensure efficient and accurate sequencing.

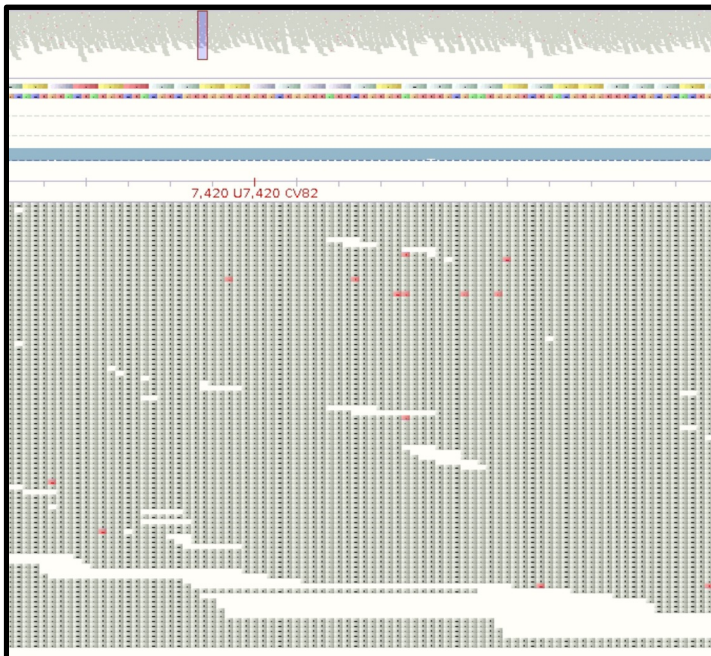
Each flow cell and chemistry combination requires its own basecalling model to ensure optimal performance and accuracy in data analysis. Specific kits may not work with all flow cells, so it is essential to match the correct kit with the appropriate flow cell.



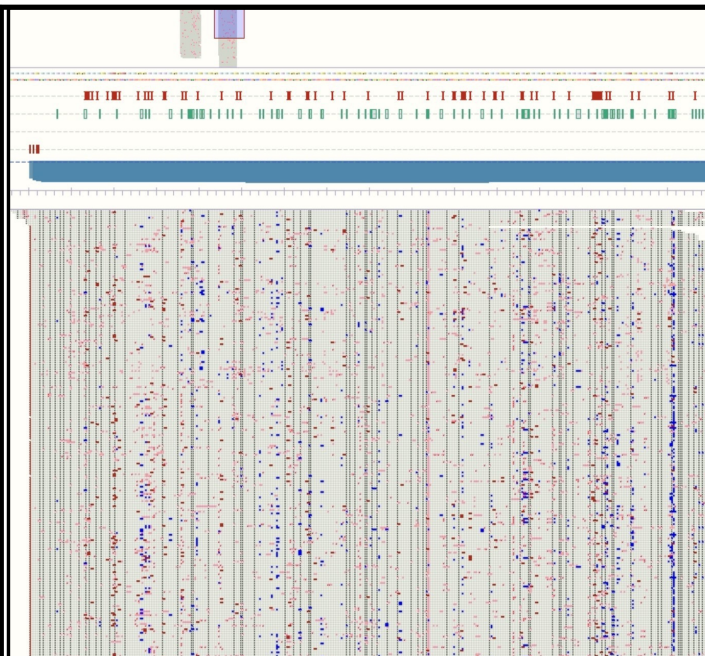
Basecalling

The output from nanopore is termed a “**squiggle**” which must be converted back into a sequence. **Dorado** has reduced the sequencing error using a more accurate basecalling model.

Illumina



Nanopore: Guppy

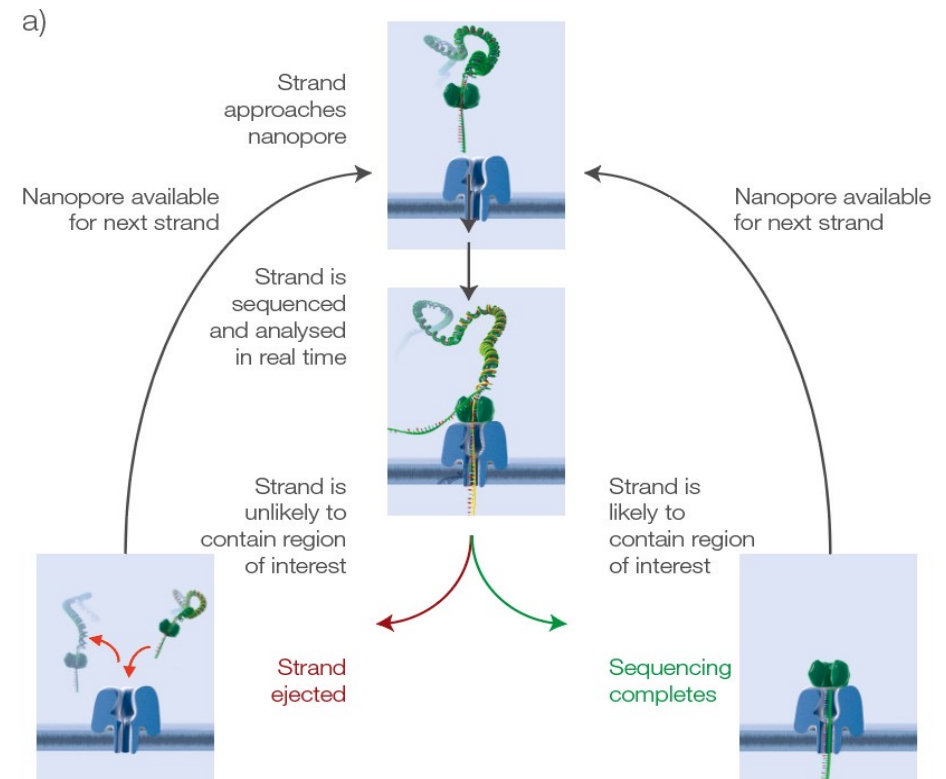


Nanopore: Dorado

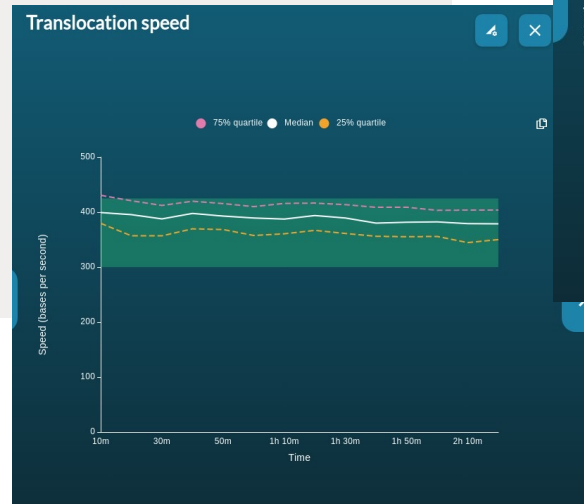
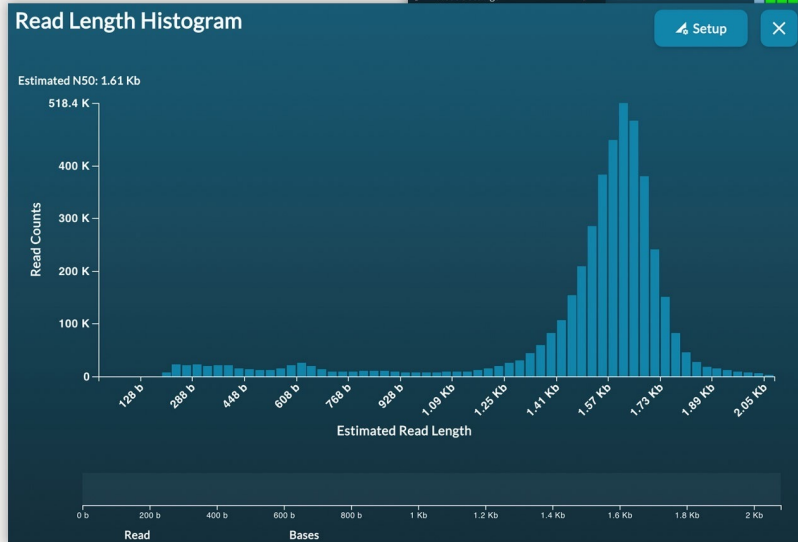
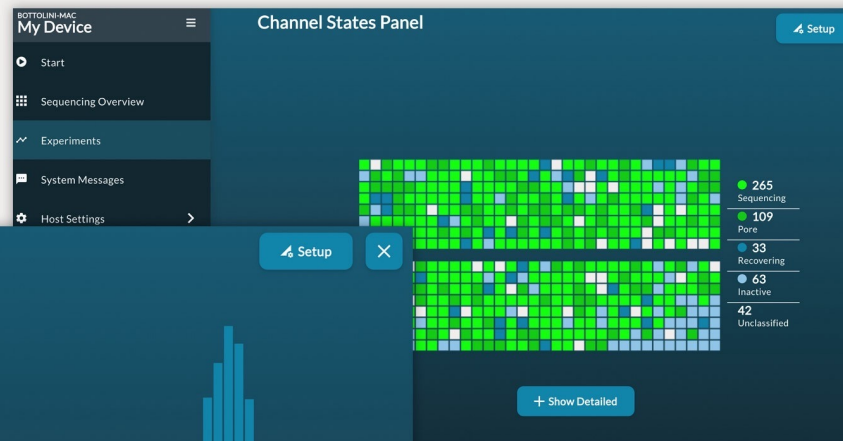


Adaptive sampling

- **Input a Reference Sequence (FASTA file):** Provide a reference genome to guide adaptive sampling.
- **Alignment:** Align data to the reference in real-time to prioritise relevant sequences.
- **Reject Non-target DNA:** Filter out non-target DNA to focus sequencing on the desired regions.
- **Selective Enrichment:** Enrich target DNA by rejecting unwanted sequences, improving sensitivity.



MinION- Real Time Sequencing



Remember to save
your PDF report in
MinKNOW



Analysis Pipeline

Basecalling

- Converts 'squiggle' to raw read data.
 - Pod5 → Fastq
- **Dorado**
 - MinKNOW currently uses Dorado for basecalling and demultiplexing.
- Fast, high accuracy and super high accuracy models.

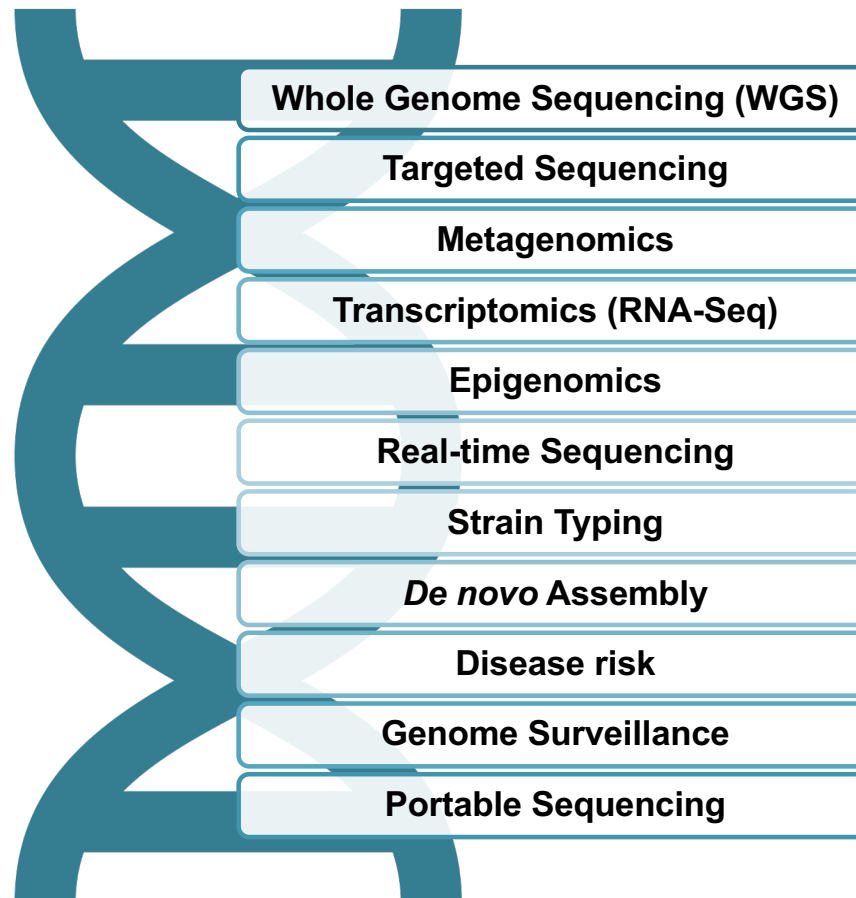
Quality Control

- **Assessment**
 - PycoQC can assess read quality and other metrics overtime.
- **Contamination**
 - Kraken2 is used to generate contamination reports.
 - Contaminants are removed using KrakenTools
- **Trimming/ Filtering**
 - Chopper

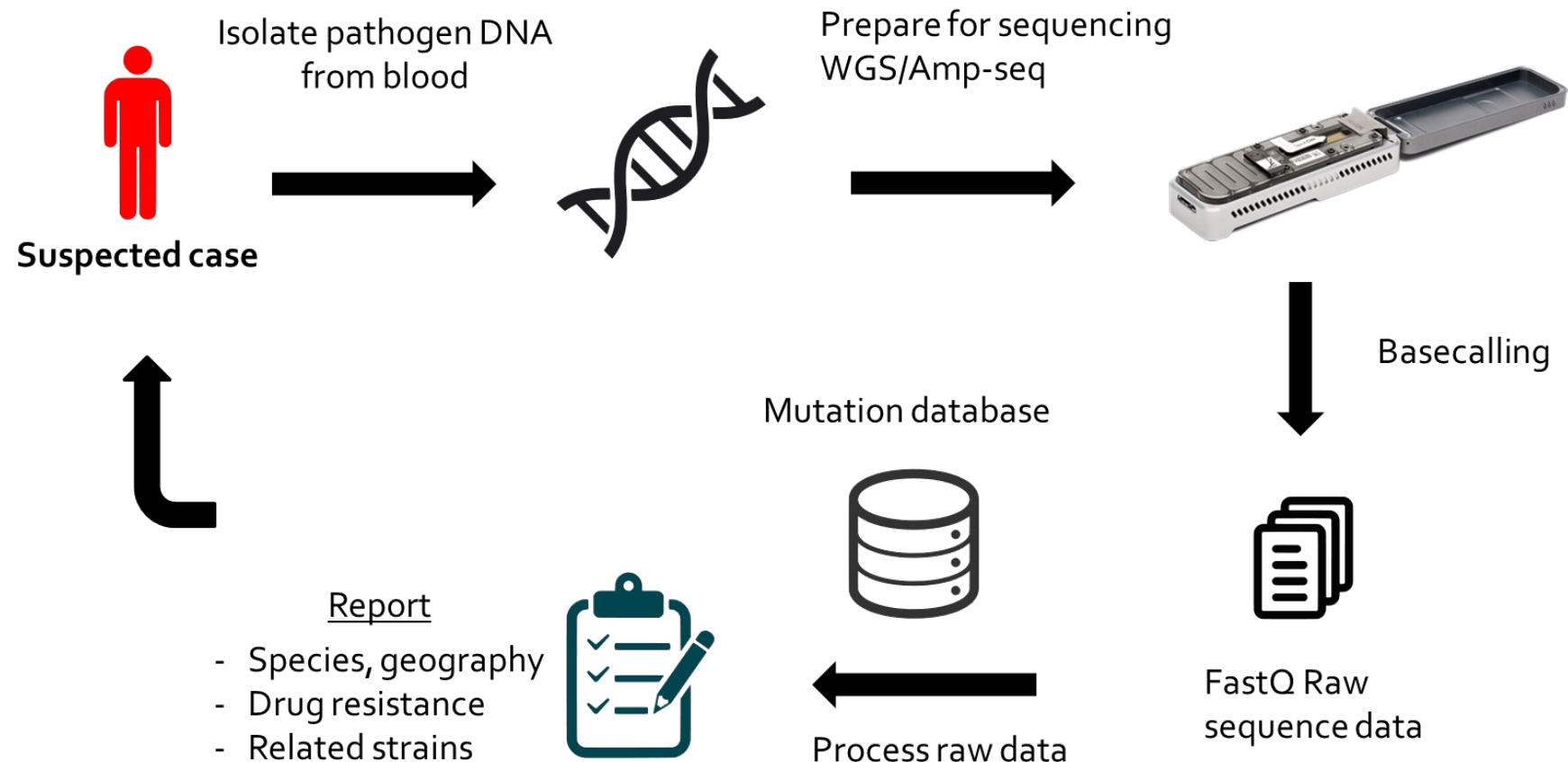
Mapping and Variant Calling

- Different tools and parameters are used in contrast to Illumina data.
- **Mapping**
 - Minimap2 (-ax map-ont)
 - BWA (-x ont2d)
- **Variant Calling**
 - Clair3
 - Freebayes
- **Assembly**
 - Flye
 - Canu

Applications of ONT sequencing

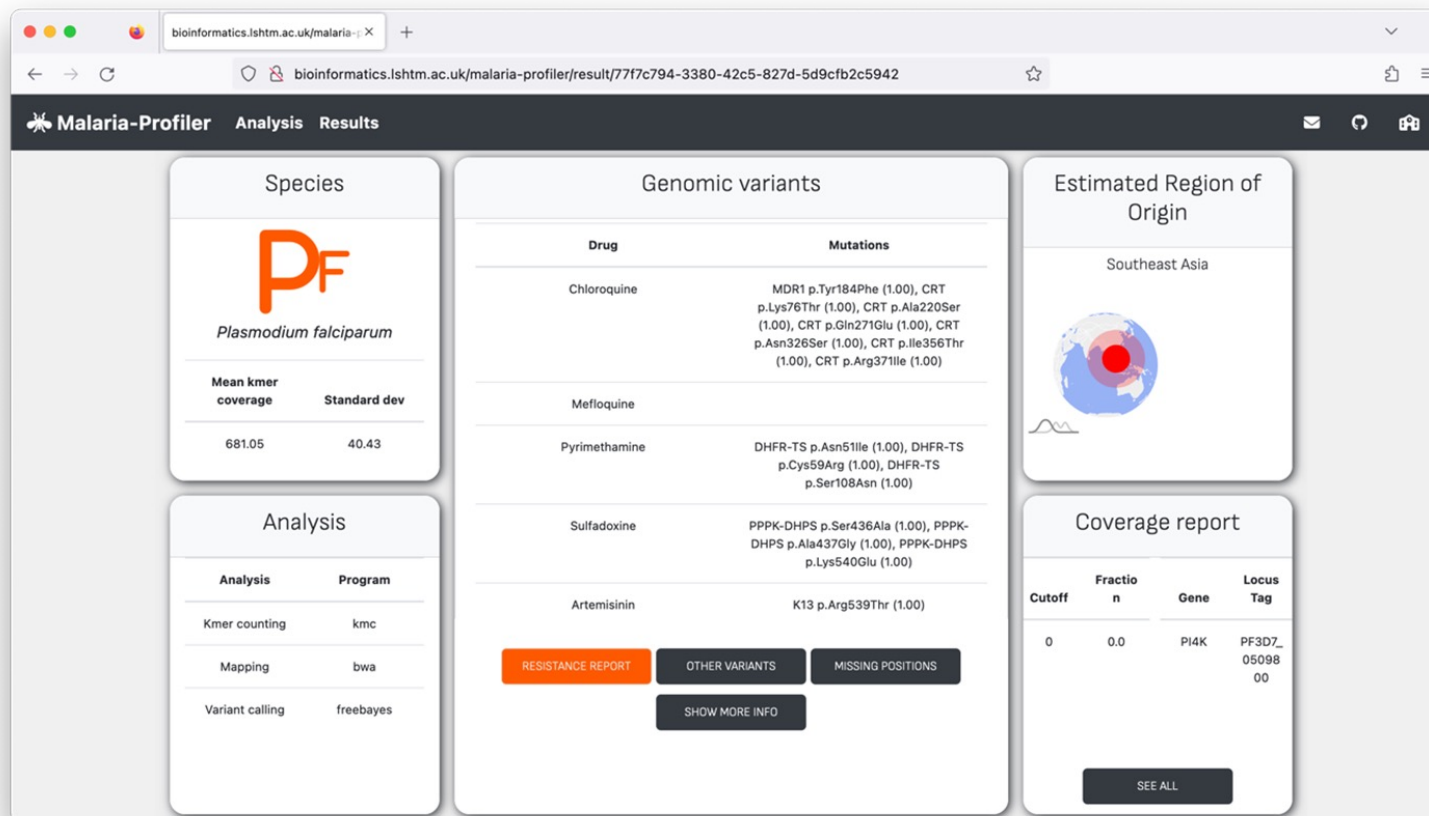


Applications: Profiling Malaria Strains

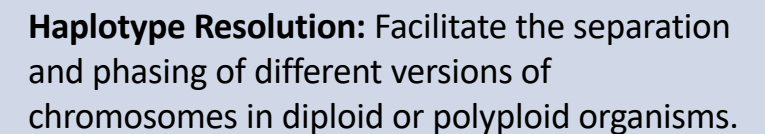
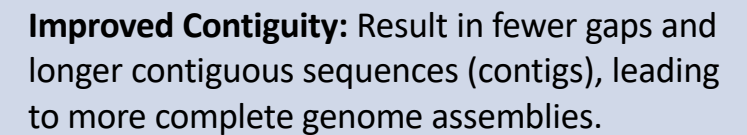
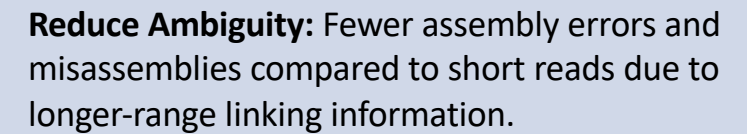
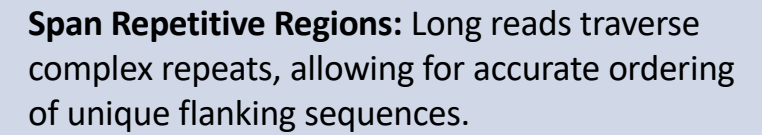
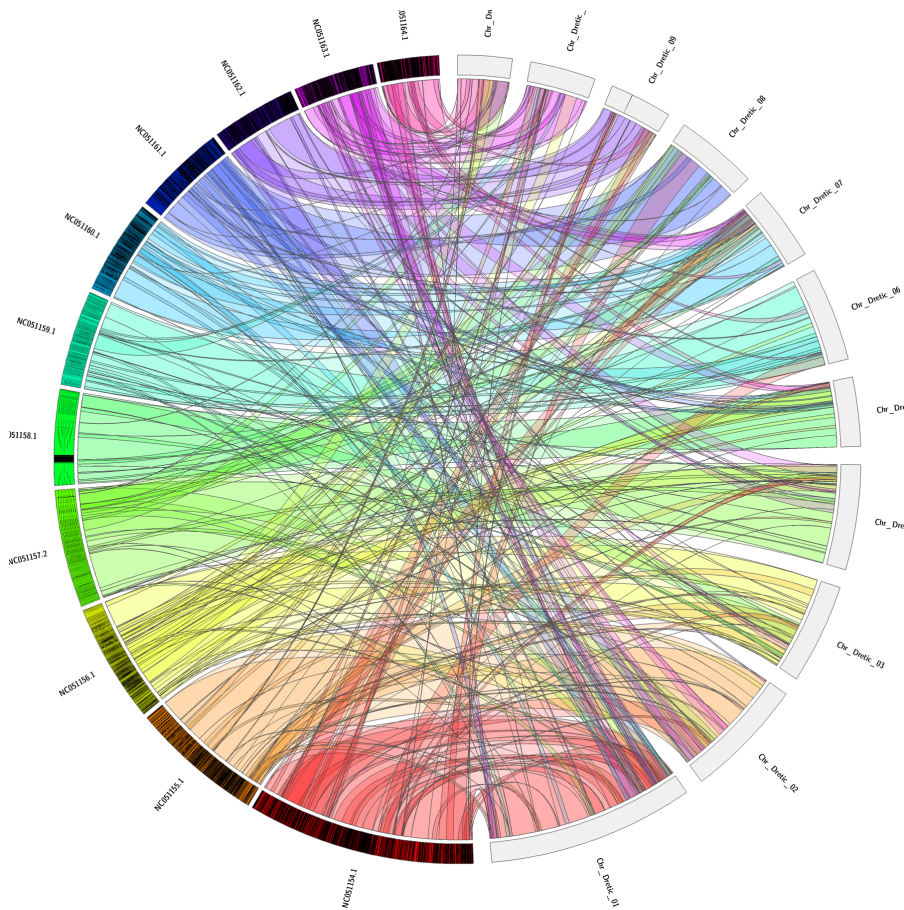


Applications: Profiling Malaria Strains

P. falciparum from Southeast Asia, with a complex drug resistance profile (example data)

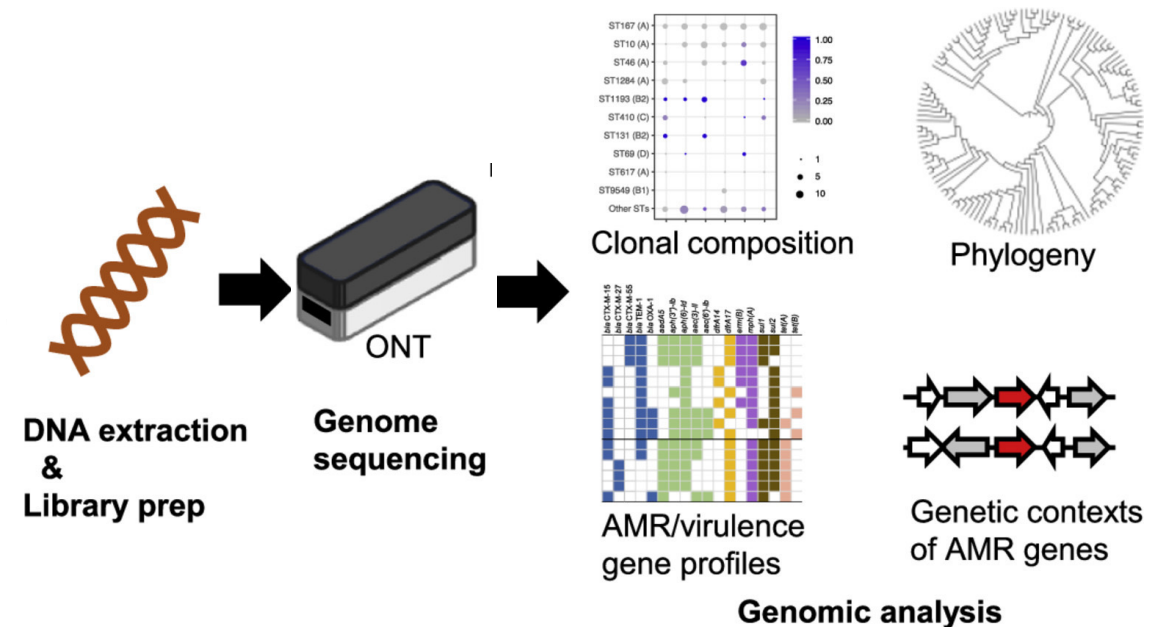


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The Nanopore Practical

- We have provided you with nanopore raw read data of *Escherichia coli* samples from a **long-term health care facility in Japan**.
- Your task is to perform a long-read assembly of this data and identify any **drug-resistance genes, virulence genes or plasmids** these samples may contain.
- Think about the quality of your data as you move through the practical and the impact this may have on your assembly.
- You will also perform a **pan-genome analysis** to compare your results to other *E. coli* samples.



Nanopore Sequencing Practical

